

Eslam A. Hugair

MLOps & Software Engineer · Data Pipelines · Production ML Infrastructure

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Summary

Software engineer with 10+ years building data-ingestion pipelines, internal tooling and the infrastructure around ML systems. Writes production Python and TypeScript services that move data through queues (BullMQ, Kafka), persist it across MongoDB / Elasticsearch / vector stores, and surface it to ML and research teams through dashboards, debugging and integrity tooling. Owns the deployment side end-to-end — Docker, CI/CD with regression-test gates, monitoring, AWS / GCP, Cloud Run — so the pipelines ship, version cleanly and roll back safely in production.

Work Experience

Mannar.sa – KSA (Remote)

09/2025 - Present · Part-Time

Sr. Software Engineer & DevOps

Saudi Arabia's first online legal-consultation platform; secure, scalable architecture for digital legal services.

- Owned end-to-end SWE: services, frontend architecture and the deployment pipeline for a high-trust legal platform.
- Owned platform security, compliance and performance work.
- Managed infrastructure, monitoring and observability for production traffic.
- Built and tuned CI/CD pipelines with regression-test gates and rollback paths to shorten the deploy cycle safely.
- Scaled the platform from pilot to thousands of active users.
- Designed and shipped the notification system with retry, dead-lettering and audit logging.
- Implemented SEO scripts and post-build meta-tag injection.
- Highlighted technologies used: **TypeScript, Node.js, React, AWS, Supabase, Firebase, MongoDB, CI/CD.**

Revic.ai – USA (Remote)

12/2022 - 09/2025

Sr. Full-Stack Engineer / Team Lead

Data-driven AI analytics platform surfacing department-level insights from heterogeneous integrations.

- Built the data-ingestion services that stream from a variety of integration tools into the analytics platform — the team's auditable intake layer.
- Designed and shipped debugging tools that helped the ML and data teams identify and resolve data-integrity issues.
- Partnered with the ML team to surface high-integrity model outputs and feedback loops into the product.
- Authored Azure Functions services for ingestion, enrichment and downstream notifications at scale.
- Worked with Cognito Search and Elasticsearch for big-data retrieval workloads.
- Built the Slack bot interface that lets clients explore use cases and act on insights.
- Promoted to team lead within 6 months for adaptability and ownership in a fast-growing team.
- Highlighted technologies used: **TypeScript, Python, Node.js, React, Azure Functions, CosmosDB, MongoDB, Elasticsearch, Cognito Search, Slack APIs.**

REPLACE-ME

02/2021 - 01/2022

REPLACE-ME (freelance / contract / sabbatical / study)

REPLACE-ME: one line on what this year was.

WeArt.io – Berlin, Germany (Remote)

02/2022 - 12/2022

Sr. Software Engineer

Development agency focused on shipping quality products.

- Recovered a real-estate project with a refactoring plan that reduced technical debt and shortened time-to-feature.
- Built features end-to-end alongside the platform refactor; expanded unit, integration and regression test coverage.
- Improved product performance, build pipelines and developer-experience around the codebase.
- Highlighted technologies used: **TypeScript, Node.js, Next.js, AWS Fargate, Jenkins, Docker, MySQL, GraphQL.**

LeadCart.io – Delaware, USA (Remote)

08/2018 - 01/2021

Chief Technology Officer (CTO)

Cart-solution startup focused on boosting conversions with targeted funnels.

- Built a custom design-system in React to back the product's design and functional requirements.
- Built utility packages of reusable functionality shared across product services.
- Owned DevOps for product services (GitHub Actions, AWS) — held 99.9% uptime across the product lifespan.
- Promoted to Team Lead within a year and CTO the year after.
- Built a standalone affiliate system to track promoter commissions — idea to production in one month.
- Built an activity-tracking service for customer events, purchases and lead capture.
- Designed a payment-integration facade fronting multiple third-party gateways.
- Helped hire and onboard the team to 5+ members; ran assessments and interviews.
- Highlighted technologies used: **Node.js, React, AWS, GitHub Actions, Netlify, Next.js, TypeScript, MongoDB, PostgreSQL.**

Zaino

05/2016 - 09/2019 · Concurrent

Team Lead, Full-Stack Engineering

Platform that augments the Google Ads customer experience with templates and campaign tooling.

- Designed and built the mobile-friendly web app for Zaino clients to manage campaigns and ship banners.
- Designed and built the API plus the Google Ads integration services that power campaign report queries.
- Built the admin console for ad-fee control, audience segments and notification management.
- Built a Socket.io customer-support dashboard for live communication with customers.
- Highlighted technologies used: **Node.js, React, AWS, PostgreSQL, Google Ads API, Socket.io.**

Selected Projects

VEngine

Engineer (team of 3) · Data pipelines + AI · 01/2026

Video knowledge engine, built with a small team of three — turns video into a searchable, clip-level knowledge graph.

- 9-stage BullMQ pipeline: FFmpeg → chunking → parallel Deepgram / Google STT → Gemini / GPT-4o summarization → embeddings → knowledge-graph + dendrogram.
- Swappable providers across STT, LLM and embeddings; A/B comparison built in.
- Per-video and per-user graphs with cross-video deduplication and SSE progress streaming.
- Deployed to Google Cloud Run; storage on local disk or GCS.

Stack: **NestJS, BullMQ, MongoDB, Redis, Deepgram, Google STT, Gemini, OpenAI, UMAP, GCS, Cloud Run, Turborepo.**

GoValidate

Co-founder · Full-stack + LLM orchestration · 01/2025 - Present

Co-founded AI-native validation platform — rough startup idea to evidence-backed verdict.

- Shipped end-to-end: NestJS API, Next.js 15 console, Docker Compose + Caddy infra.

- BullMQ multi-agent pipeline mining Reddit signals through multi-provider LLMs (OpenAI + Anthropic via LangChain).
- Hybrid vector + keyword search on Qdrant; 0–3 source-credibility scoring across academic, government and press.

Stack: **Next.js 15, NestJS, TypeScript, BullMQ, MongoDB, Qdrant, OpenAI, Anthropic, LangChain, Docker, Caddy.**

<https://govalidate.dev>

QuiQ

Lead engineer · DevOps · 01/2025 - Present

Lead engineer on an internal SAT-content review platform — feedback, activity tracking, RBAC, migration tooling.

- Migrated backend from Netlify Functions to a long-lived Express API for predictable load and lower latency.
- Built feedback collection, activity / request tracking, RBAC (super-admin / admin / user) and the invitation flow.
- Owned the migration tooling piping curated QuiQ content into Puef.
- Owned DevOps end-to-end: environments, CI, monitoring, staging / production split.

Stack: **React, TypeScript, Node.js, Express, MongoDB, Auth0, Zod, Netlify.**

<https://quiq.netlify.app/>

Puef.ai

Contributor · LLM endpoints + frontend · 01/2025 - Present

AI SAT-learning platform — adaptive planning, review scheduling and content per learner.

- Hardened the LLM generation endpoints and built the 'fixer' flows that repair invalid or unclear outputs.
- Worked across the multi-provider LangChain layer (OpenAI / Anthropic / Gemini / Deepseek).
- Frontend work and the integration glue with the QuiQ review pipeline.

Stack: **Python, Flask, LangChain, OpenAI, Anthropic, Gemini, Deepseek, MongoDB, React, TypeScript, Railway.**

<https://puef.ai>

Also built: Story-Teller (multi-agent story-to-video: GPT-4o, DALL-E 3, ElevenLabs) - Shorts-Gen (long-form video to stylized shorts: Python, FFmpeg, style transfer) - Denture Mesh + Content Engine (Open3D / NumPy mesh annotation with an LLM content layer).

Technical Skills

- **Data & MLOps:** Production Python (pandas, NumPy, Open3D, pytest); data-ingestion pipelines; data quality, schema validation and integrity checks; queue-based orchestration (BullMQ; concepts transfer to Airflow / Prefect); vector stores (Qdrant, Pinecone); embeddings; Deepgram / Google STT; debugging + integrity tooling for ML teams.
- **Deployment & Reliability:** CI/CD with regression-test gates, containerized services (Docker, Docker Compose), blue/green and rollback-friendly releases, monitoring & logging, observability for data pipelines, on-call experience.
- **Cloud & Infra:** AWS (S3, IAM, Fargate), Google Cloud (Cloud Run, GCS), Azure Functions, GitHub Actions, Jenkins, Railway.
- **Languages & Frameworks:** Python, TypeScript, Node.js, NestJS, Flask, FastAPI-style services, React, Next.js.
- **Storage & Search:** MongoDB, PostgreSQL, MySQL, CosmosDB, Redis, Elasticsearch, Kafka, Qdrant.
- **Testing & Quality:** Unit + integration testing, regression suites, contract tests for ingestion pipelines, data-validation harnesses, code reviews.
- **LLM tooling (working knowledge):** Multi-provider LLM gateways (OpenAI, Anthropic, Gemini) via LangChain; RAG; prompt fixers / evaluation loops; agentic pipelines.
- **Practices:** Internal tooling and dashboards for research teams, custom annotation platforms, end-to-end ownership, mentoring.

Education

B.Sc. Software Engineering — REPLACE-ME University

02/2012 - 06/2017

Languages

- Arabic (native) · English (advanced)